# Mixes, covers and faithfulness: the three-source measurement boundary

Human Compression project

Input proposal; distinct review required

This proposal adds *Faithfulness in maniplexes* (Cunningham, Mochán, Montero, Potočnik and Toledo, 2026) to the existing F005/L016 topic. It fixes whole documents and records their text lengths. It evaluates no compression factor and makes no new mathematical or formalization claim.

| Component | Pages | Treatment |
|---|---|---|
| F005 source | 21 | Exact accepted edition and measurements |
| L016 source | 12 | Exact accepted edition and measurements |
| F007 source | 29 | Complete journal edition, including references |
| F005 common paper | 5 | Entire paper required |
| L016 common paper | 4 | Entire paper required |
| F007 common paper | 4 | Entire paper required |
| F005 applications | 8+4 | Both complete papers |
| L016 application | 6 | Complete paper |
| F007 application and companion | 5+4 | Both complete papers |

The written replacement boundary contains all three common papers once and all five application papers. There is no deduction for overlap and no free earlier common theory. In particular, the four-page rank-three companion is an application component; its short last page remains included.

The primary coordinate is the number of non-whitespace characters in the *complete raw PDF extraction*, normalized to NFC. It was fixed before new counts. The same five units are recorded under NFC and NFKC for raw, furniture-selected and glyph-calibrated selected text: whitespace tokens, alphanumeric runs, non-whitespace characters, collapsed-whitespace characters and collapsed-whitespace UTF-8 bytes. These are thirty coordinates per PDF, not different claims of mathematical complexity.

All 330 accepted PDF cells and 130 raw-file cells from F005/L016 are preserved exactly. Four new PDFs contribute 120 cells; four new raw files contribute 40 cells. The full inventory therefore has 450 PDF cells and 170 raw-file cells. Separate historical program appendices remain inventoried but are not added to the written manuscripts or counted twice with their raw files.

The new primary non-whitespace counts are 63,311 for the F007 source, 10,501 for its common paper, 13,841 for its application and 8,650 for its companion. No quotient of these numbers is evaluated here.

**Programs and formal evidence.** The complete topic retains two required finite certificates, one each from F005 and L016. They remain mathematical premises and are recorded in full, with the existing two-page required-code representation as an alternative. Six optional validation programs are recorded separately: the four inherited files and F007's two finite diagnostics. The latter are not premises for its uniform proofs. Optional checking is not a prerequisite for a conditional written comparison. No software-inclusive factor is supplied: corresponding historical source software costs remain unknown.

Six whole operational drivers and three whole Lean files are separately inventoried. F007's companion checker also performs its own rendering; it is charged once as a complete diagnostic file, with that second role stated. The Lean inventory contains twelve scoped declarations (two F005, four L016 and six F007), not an end-to-end formalization of any source. No scientific program or Lean theorem is rerun by this input proposal.

**Actual glyphs and selection.** The new source has twenty actual one-byte font dictionaries and 443 used encoded classes. Dictionary identity, effective encoding, embedded stream, operator, byte, page and exact extraction context are bound independently. Thirty-four exceptional outlines were viewed. Eighteen classes, occurring 1,351 times, receive calibrated substitutes: six Greek capitals, six script capitals, four blackboard-bold classes, an envelope and the visible mapsto component. Raw extraction is unchanged. Existing correctly decoded symbols, including the phi form and the combining negation stroke, remain literal.

The source's publisher footer extracts as `123`; its actual glyphs are a horse emblem, an empty outline and the word Springer. All three characters remain in raw text. Selected variants remove the exact footer and running headers only. The first-page journal identification, author addresses, references, acknowledgements and licence remain included. Own selected variants remove only isolated page-number footers. The publisher's remaining logos, the six numbered source figures and all other graphical structure retain unknown complexity; extracted labels remain.

**Symmetry and scope.** All 29 F007 source pages and all three new scientific texts were screened for executable listings. There are none. Displayed group presentations and adjacency formulas are retained mathematical statements, not program masks. The accepted F005/L016 screening is reused with exact bindings. Required code and optional tests remain separate inventories on both sides; unprovided historical software is not assigned zero cost.

The scientific boundary remains the accepted corrected F005 and L016 contracts plus F007's 48 numbered results and six additional dispositions. F007's unnumbered chain-class paragraph on page 25 is not recovered; Problems 1 and 2 remain open here. Problem 3 is treated through natural reconstruction, and arbitrary isomorphism for finite posets with connected flag graph; no general infinite arbitrary-isomorphism or priority claim is made. The input author supplied F007's distinct science review; root adopted that review. This measurement proposal still requires a different reviewer.